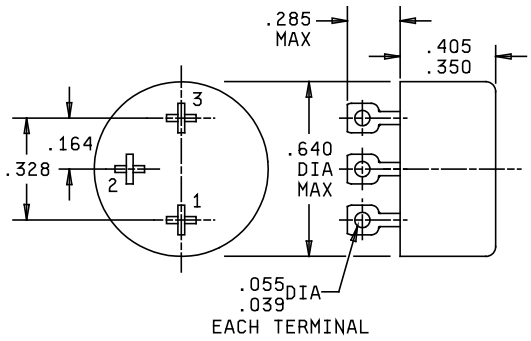
INCH-POUND
MIL-PRF-24236/24F
6 June 2022
SUPERSEDING
MIL-PRF-24236/24E
5 April 2006

PERFORMANCE SPECIFICATION SHEET

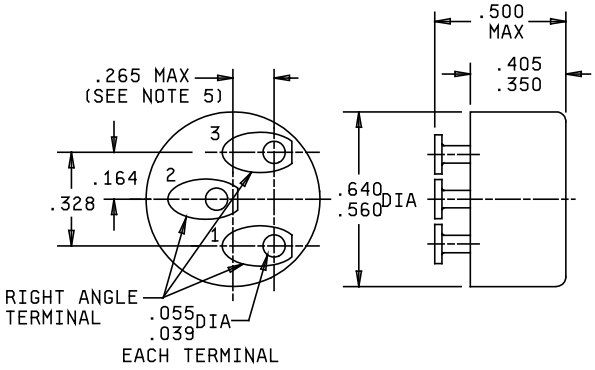
# SWITCHES, THERMOSTATIC, (BIMETALLIC), SUBMINIATURE, TYPE I, HERMETICALLY SEALED, SINGLE POLE, DOUBLE THROW (SPDT), 2 AMPERES AND LOW LEVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-PRF-24236.




Configuration A (see note 6)

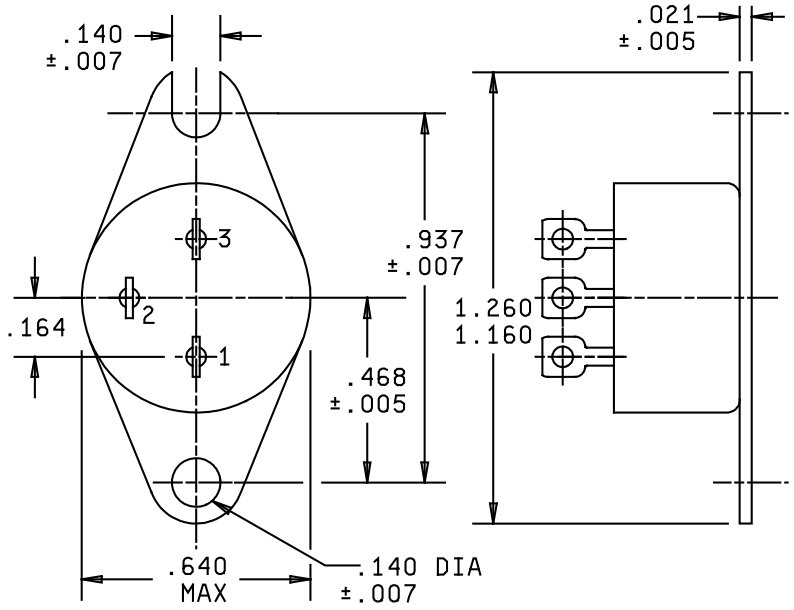



Configuration B (see note 6)

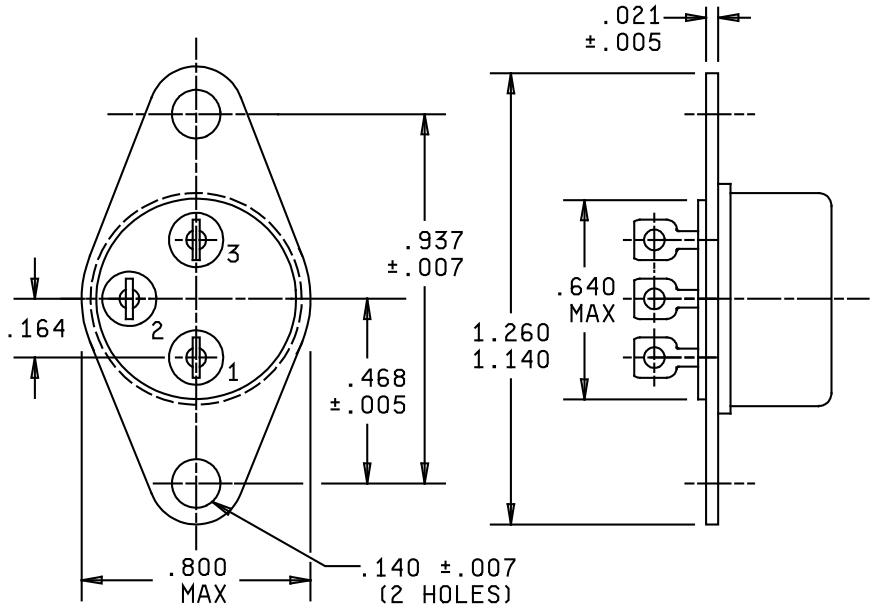
FIGURE 1. Switches.

AMSC N/A
FSC 5930
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.




Configuration A1

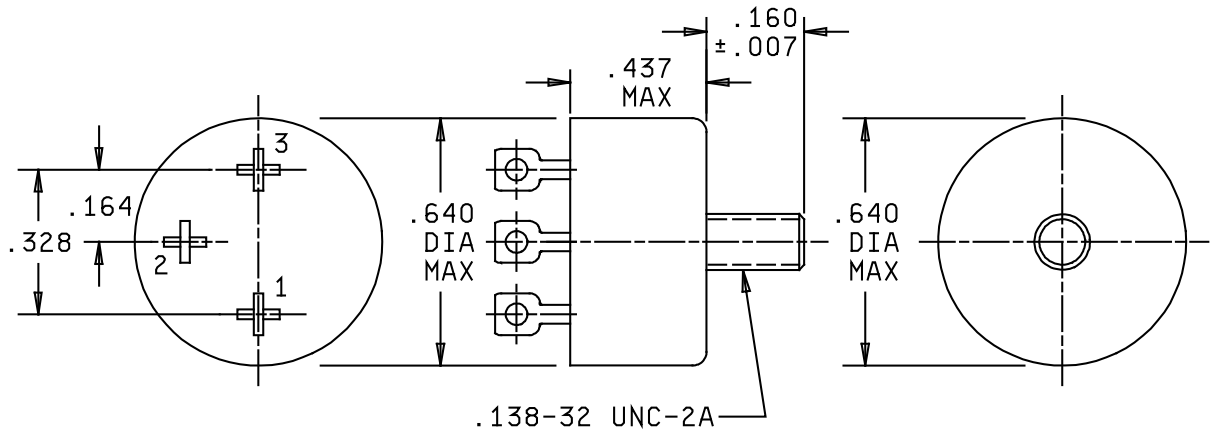


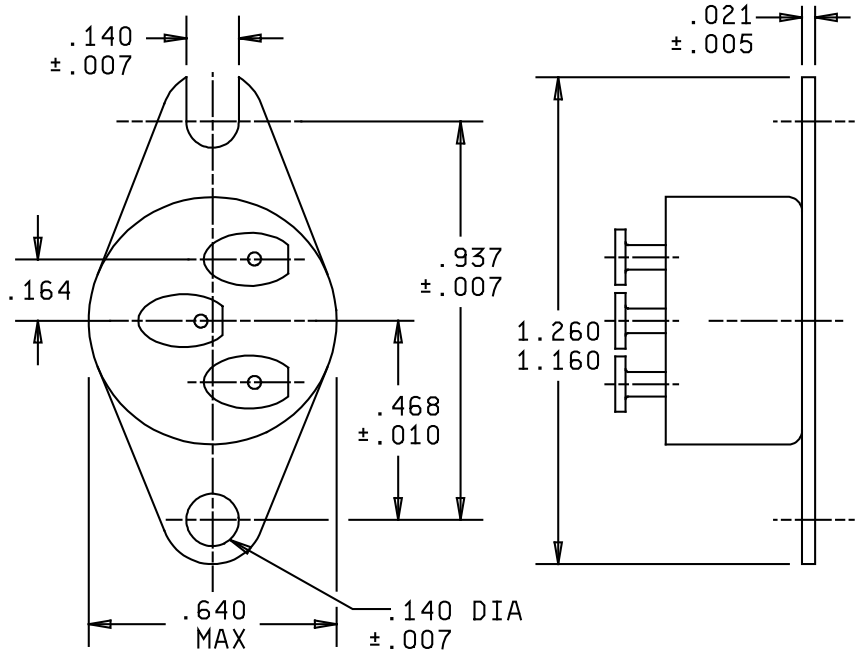

Configuration A2

FIGURE 1. Switches - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.



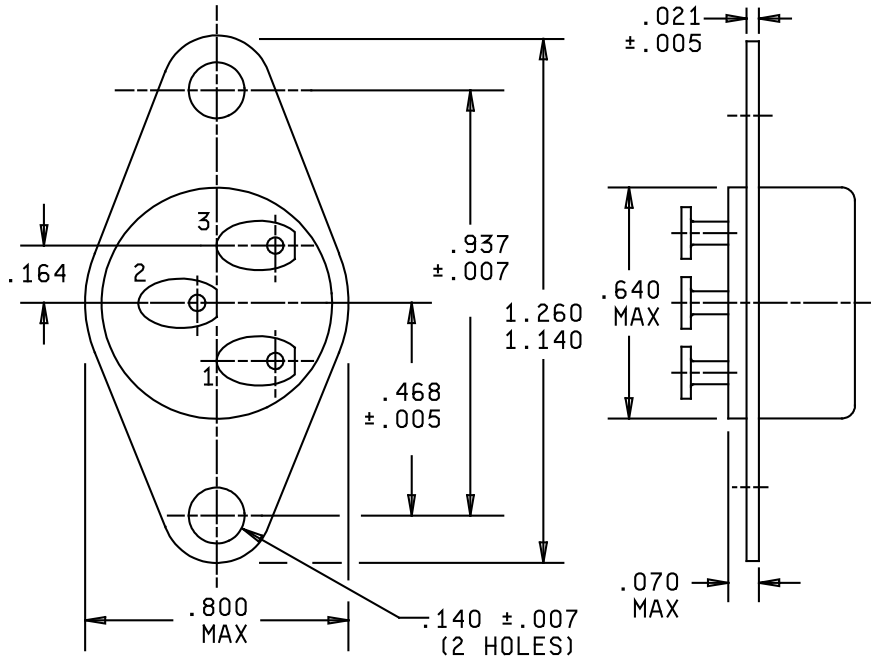

Configuration B1

Configuration A3 (stud mounted)

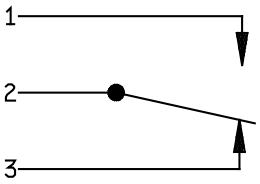
FIGURE 1. Switches - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.




Configuration B2




WIRING DIAGRAM

Below low temperature setting, circuit 1-2 is open.
Above high temperature setting, circuit 2-3 is open.

NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerance is ±.015.
2. Exact shape of switch and terminals are optional provided dimensions specified are not exceeded.
3. This dimension not to exceed extended envelope of the switch.
4. Mounting brackets are not to be furnished.
5. Configurations A1, A2, and A3 use basic switches of configuration A.
   Configurations B1, B2, and B3 use basic switches of configuration B.

FIGURE 1. Switches - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Operating temperature range: -65°F to 400°F.

Class: Class 4, except 80,000 feet and vibration 10 - 2,000 Hz, 10G.

Mounting: See figure 1.

Weight: Not to exceed .025 pound.

Creepage: Applicable.

Operating temperature and tolerance: See table II and table III.

Torque (switch mounting): 15 foot-pounds.

Electrical ratings: See table IV.

Endurance test: See table IV.

Low level: When specified, see PIN information.

QUALIFICATION:

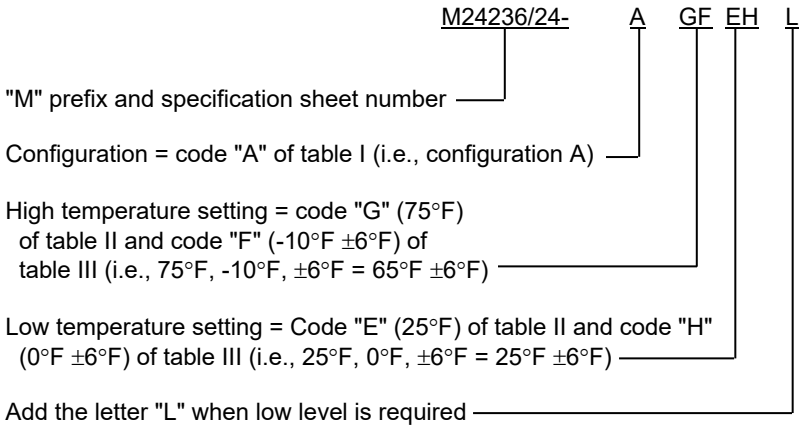
Single submission: Restricted to switch submitted.

Group submission: See table V.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number; a dash (-); and a five- or six-letter code. The five-letter code identifies the configuration (code from table I); high temperature setting to the nearest 25°F (code from table II); high-temperature setting to the nearest 5°F and applicable tolerance (code from table III); low-temperature setting to the nearest 25°F (code from table II); and low-temperature setting to the nearest 5°F and applicable tolerance (code from table III). The six-letter code used in the following example identifies a switch of configuration A (figure 1) which actuates at 65°F ±6°F and deactuates at 25°F ±6°F and has low level capabilities.

Note: Refer to table VI to avoid generating a PIN that cannot be acquired.

EXAMPLE:






Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.

TABLE I. Configuration.

| | Configuration | | | | | | |
|---|---|---|---|---|---|---|---|
| Code - - - - | A | A1 | A2 | A3 | B | B1 | B2 |
| | A | B | C | D | E | F | G |

TABLE II. High temperature setting to the nearest 25°F.

| Code | Temperature (°F) | Code | Temperature (°F) | Code | Temperature (°F) |
|---|---|---|---|---|---|
| A - - - - | -75 | H - - - - | 100 | Q - - - - | 275 |
| B - - - - | -50 | J - - - - | 125 | R - - - - | 300 |
| C - - - - | -25 | K - - - - | 150 | S - - - - | 325 |
| D - - - - | 0 | L - - - - | 175 | T - - - - | 350 |
| E - - - - | 25 | M - - - - | 200 | U - - - - | 375 |
| F - - - - | 50 | N - - - - | 225 | V - - - - | 400 |
| G - - - - | 75 | P - - - - | 250 | | |

TABLE III. Temperature setting to the nearest 5°F.

| | Unit | | | | | |
|---|---|---|---|---|---|---|
| | -10 | -5 | 0 | +5 | +10 | Tolerance |
| Code - - - - - - - - - - | A | B | C | D | E | ±4°F |
| Code - - - - - - - - - - | F | G | H | J | K | ±6°F |
| Code - - - - - - - - - - | L | M | N | P | Q | ±8°F |
| Code - - - - - - - - - - | R | S | T | U | V | ±10°F |
| Code - - - - - - - - - - | W | X | Y | Z | 1 | ±12°F |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.

TABLE IV. Electrical ratings.

| Load | Altitude | | | Life cycles |
|---|---|---|---|---|
| | Sea level | | 80,000 feet | |
| | 28 V | 115 V, 60 Hz | 28 V | |
| | (amperes) | (amperes) | (amperes) | |
| Resistive | 2.0 | 2.0 | 2.0 | 100,000 |
| inductive | 1.0 | 1.0 | 1.0 | 100,000 |
| Lamp | 0.5 | 0.5 | 0.5 | 100,000 |
| Low level ratings | | | | |
| | 30 millivolts dc 1/ | | | |
| | 10 milliamperes | | | 250,000 |

1/ Or 30 millivolts peak ac.

TABLE V. Extent of qualification.

| Configuration | Number of samples required | Tests | Qualifies |
|---|---|---|---|
| A1 | All in accordance with qualification table of MIL-PRF-24236 | Complete in accordance with qualification inspection of MIL-PRF-24236 | All |
| A, A2, B, B1, B2 | Two each | Visual and mechanical | |
| A3 | Two each | Visual and mechanical, torque, vibration, and shock | |

TABLE VI. Available differential range and recommended tolerance.

| Operating temperature range (°F) | Available differential range | | Available tolerance | |
|---|---|---|---|---|
| | Min (°F) | Max (°F) | Low (°F) | High (°F) |
| -65 to -1 | 25 | 200 | ±8 | ±10 |
| 0 to 300 | 15 | 200 | ±6 | ±8 |
| 301 to 350 | 15 | 200 | ±8 | ±10 |
| 351 to 400 | 25 | 200 | ±12 | ±12 |

NOTE: Tolerances not listed in table VI shall be coordinated with manufacturers listed on the QPL for this specification sheet before specifying a particular "M" number. The fact that tolerances can be coded does not necessarily mean that it can be manufactured or acquired.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.

Referenced Documents:
MIL-PRF-24236

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2022-046)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:32Z
Check the source to verify that this is the current version before use.